In [1]:
```
import pandas as pd
import numpy as np
df= pd.read_csv("Telco Customer Churn.csv")
df.head(5)
```

Out[1]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	Multipl
0	7590-VHVEG	Female	0	Yes	No	1	No	No
1	5575-GNVDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	No
4	9237-HQITU	Female	0	No	No	2	Yes	

5 rows × 21 columns



In [2]:
```
df.columns
```

Out[2]:
```
Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',
       'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
       'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',
       'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',
       'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],
      dtype='object')
```

In [3]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   customerID        7043 non-null   object 
 1   gender            7043 non-null   object 
 2   SeniorCitizen     7043 non-null   int64  
 3   Partner           7043 non-null   object 
 4   Dependents        7043 non-null   object 
 5   tenure            7043 non-null   int64  
 6   PhoneService      7043 non-null   object 
 7   MultipleLines     7043 non-null   object 
 8   InternetService   7043 non-null   object 
 9   OnlineSecurity    7043 non-null   object 
 10  OnlineBackup      7043 non-null   object 
 11  DeviceProtection  7043 non-null   object 
 12  TechSupport       7043 non-null   object 
 13  StreamingTV       7043 non-null   object 
 14  StreamingMovies   7043 non-null   object 
 15  Contract          7043 non-null   object 
 16  PaperlessBilling  7043 non-null   object 
 17  PaymentMethod     7043 non-null   object 
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object 
 20  Churn             7043 non-null   object 
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

In [4]:
```
df.shape
```

Out[4]:
```
(7043, 21)
```

In [5]:
```
df.isnull().sum()
```

Out[5]:
```
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
```

In [6]:
```
print(df.duplicated().sum())
```

```
0
```

In [7]:
```
df.dtypes
```

Out[7]:
```
customerID           object
gender               object
SeniorCitizen         int64
Partner              object
Dependents           object
tenure                int64
PhoneService         object
MultipleLines        object
InternetService      object
OnlineSecurity       object
OnlineBackup         object
DeviceProtection     object
TechSupport          object
StreamingTV          object
StreamingMovies      object
Contract             object
PaperlessBilling     object
PaymentMethod        object
MonthlyCharges      float64
TotalCharges         object
Churn                object
dtype: object
```

In [8]:
```
df[['tenure','MonthlyCharges']].describe()
```

Out[8]:

	tenure	MonthlyCharges
count	7043.000000	7043.000000
mean	32.371149	64.761692
std	24.559481	30.090047
min	0.000000	18.250000
25%	9.000000	35.500000
50%	29.000000	70.350000
75%	55.000000	89.850000
max	72.000000	118.750000

In [9]:
```
df.drop_duplicates()
```

Out[9]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	Mul
0	7590-VHVEG	Female	0	Yes	No	1	No	
1	5575-GNVDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	
4	9237-HQITU	Female	0	No	No	2	Yes	
...	...	...	...	...	...	...	...	
7038	6840-RESVB	Male	0	Yes	Yes	24	Yes	
7039	2234-XADUH	Female	0	Yes	Yes	72	Yes	
7040	4801-JZAZL	Female	0	Yes	Yes	11	No	
7041	8361-LTMKD	Male	1	Yes	No	4	Yes	
7042	3186-AJIEK	Male	0	No	No	66	Yes	

7043 rows × 21 columns



In [10]:
```
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
print("Nulls found after fixing type:", df['TotalCharges'].isnull().sum())
```

```
Nulls found after fixing type: 11
```

In [11]:
```
df['TotalCharges'] = df['TotalCharges'].fillna(0)
print("Nulls after fill:", df['TotalCharges'].isnull().sum())
```

```
Nulls after fill: 0
```

In [12]:
```
print("Duplicates before:", df.duplicated().sum())
df = df.drop_duplicates()
print("Duplicates after:", df.duplicated().sum())
```

```
Duplicates before: 0
Duplicates after: 0
```

In [14]:
```
text_cols = df.select_dtypes(include='object').columns.drop('customerID')
for col in text_cols:
    print(col, ":", df[col].unique())
```

```
gender : ['Female' 'Male']
Partner : ['Yes' 'No']
Dependents : ['No' 'Yes']
PhoneService : ['No' 'Yes']
MultipleLines : ['No phone service' 'No' 'Yes']
InternetService : ['DSL' 'Fiber optic' 'No']
OnlineSecurity : ['No' 'Yes' 'No internet service']
OnlineBackup : ['Yes' 'No' 'No internet service']
DeviceProtection : ['No' 'Yes' 'No internet service']
TechSupport : ['No' 'Yes' 'No internet service']
StreamingTV : ['No' 'Yes' 'No internet service']
StreamingMovies : ['No' 'Yes' 'No internet service']
Contract : ['Month-to-month' 'One year' 'Two year']
PaperlessBilling : ['Yes' 'No']
PaymentMethod : ['Electronic check' 'Mailed check' 'Bank transfer (automatic)'
 'Credit card (automatic)']
Churn : ['No' 'Yes']
```

In [15]:
```
for col in ['tenure', 'MonthlyCharges', 'TotalCharges']:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    low = Q1 - 1.5 * IQR
    high = Q3 + 1.5 * IQR
    outliers = df[(df[col] < low) | (df[col] > high)]
    print(col, "-> outliers found:", len(outliers))
```

```
tenure -> outliers found: 0
MonthlyCharges -> outliers found: 0
TotalCharges -> outliers found: 0
```

In [21]:
```
df['customerID'] = df['customerID'].astype(str)
df['TotalCharges'] = df['TotalCharges'].astype(float)
df['MonthlyCharges'] = df['MonthlyCharges'].astype(float)
df['tenure'] = df['tenure'].astype(int)
print(df.dtypes)
```

```
customerID           object
gender               object
SeniorCitizen         int64
Partner              object
Dependents           object
tenure                int64
PhoneService         object
MultipleLines        object
InternetService      object
OnlineSecurity       object
OnlineBackup         object
DeviceProtection     object
TechSupport          object
StreamingTV          object
StreamingMovies      object
Contract             object
PaperlessBilling     object
PaymentMethod        object
MonthlyCharges      float64
TotalCharges        float64
Churn                object
dtype: object
```

In [22]:
```
summary = pd.DataFrame({
    'Check': ['Rows', 'Nulls (isnull)', 'Duplicates', 'TotalCharges dtype'],
    'Before': [7043, 0, 0, 'text (object)'],
    'After': [df.shape[0], df.isnull().sum().sum(), df.duplicated().sum(), df['Tota
})
print(summary)
```

```
                Check         Before    After
0                Rows           7043     7043
1      Nulls (isnull)              0        0
2          Duplicates              0        0
3  TotalCharges dtype  text (object)  float64
```

In [23]:
```
df.to_csv('telco_churn_cleaned.csv', index=False)
print("Saved. Final shape:", df.shape)
```

```
Saved. Final shape: (7043, 21)
```

In [ ]: